Yongkang Li

Mail: y.li7@uva.nl **Phone:**(+31) 0635630756 **Home Page**

Research Interests: NLP, Information Retrieval, Robust Language Model Embeddings, Recommender systems

Education

- **University of Amsterdam, Informatics Institute**
  Lab: *Information Retrieval Lab* **PhD**, Oct 2023–Jun 2027(expected), Amsterdam, The Netherlands
  Supervisors: Prof. dr. Evangelos Kanoulas and dr. Panagiotis Eustratiadis
- **Southern University of Science and Technology, Department of Computer Science and Engineering**
  Lab: *SUSTech-UTokyo Joint Research Center on Super Smart City* **Master**, 2020–2023, Shenzhen, China
  Supervisors: Prof. dr. Xuan Song and Prof. dr. Zipei Fan
- **Beijing University of Posts and Telecommunications, School of Information and Communication Engineering**
  Lab: *Pattern Recognition and Intelligent Systems Laboratory* **Bachelor**, 2016–2020, Beijing, China

Main Research Projects

- **Understanding and Enhancing Robustness in Dense Information Retrieval**
  Embedding-based retrieval is widely deployed, yet dense retrievers remain fragile under distribution shifts and adversarial manipulation. My PhD studies robustness along two axes: *generalizability* (consistency across tasks, corpora, and query distributions) and *stability* (resilience to benign perturbations and malicious attacks). We follow a two-phase agenda: (RQ1) *Understanding* failure modes in modern retrievers, especially LLM-based dense retrievers; and (RQ2) *Enhancing* robustness via principled defenses beyond standard augmentation, including robust adaptation and training objectives.
  In our recent **ACL Rolling Review submission** (Jan 2026), we systematically evaluate SOTA LLM-based retrievers and show that, despite strong in-distribution performance, they remain vulnerable to *semantics-preserving* query edits, revealing robustness gaps not closed by scaling alone. Earlier, our **ECIR'2025** and **SIGIR'2025** work on corpus poisoning contributed efficient attack methods and practical adversarial training, providing a foundation for these broader robustness studies and mitigations. More recently, in our **SIGIR'26** work, we extend this line toward *efficient post-hoc control of embedding representations*, studying how spectral redundancy can be reduced to obtain more compact embeddings while preserving retrieval effectiveness.
- **Heterogeneous Graph Learning for Location-based Social Network Friend Recommendation**
  This research, part of my master's studies, focuses on friend recommendation in location-based social networks (LBSN) using user trajectory data. We explored the impact of social network relationships on human mobility, particularly long-distance travel. Our approach used heterogeneous multi-graphs and hypergraphs, with novel neural network models for learning and recommendation. The solution achieved state-of-the-art results on multiple real-world datasets and has been published in **TKDD**, **CIKM**, and **WWWJ**.

Industry Experience

Temu Inc. - Research and Development Department Shanghai, China
Machine Learning Algorithm Engineer Jul 2023–Sep 2023

- **Conversational Intent Recognition**
  On e-commerce platforms, customers typically have specific intents when communicating with customer service, such as requesting a return, a refund, or inquiring about shipping status. The goal of this project is to assist customer service in quickly identifying the user's intent during conversations, enabling better and faster service. We integrate deep learning methods with traditional feature-matching techniques, combining customer purchase history, current order status, and cleaned dialogue data. This approach has enabled us to develop an automated classification and statistical system that assists customer service agents and supports data analysis for other internal business departments.

Pinduoduo Inc. - Research and Development Department Shanghai, China
Machine Learning Internship Jul 2022–Sep 2022

- **Knowledge Graph Embedding**
  Investigate the development history and typical technologies of knowledge graph embedding models (such as TransE, TransR, DisMult, ComplEx, KG-Bert, RotatE, and PairRE), implement the corresponding codes and do academic talk with colleagues across the department.

- **Aspect-based Sentiment Analysis**
  Based on the e-commerce review data, I implemented the aspect-based sentiment analysis module via PaddleNLP, which extracts a quadruple like (attribute, opinion, sentiment, attribute category) from consumer review data. And this module has been applied to the company business.

Technical Skills

English:
- TOEFL-107, Reading 29, Listening 29, Speaking 23, Writing 26

Programming:
- Python, C++, Linux, Git; strong software engineering and coding practices.
- ML/DS stack: PyTorch, Pandas, NumPy, Scikit-learn, TensorFlow.
- HPC: proficient with Slurm for job scheduling, GPU/CPU resource allocation, and large-scale experimentation.

Selected Publications

Papers - First Author or Co-first Author (* indicates equal contribution)

- **Yongkang Li**, Panagiotis Eustratiadis, Yixing Fan, and Evangelos Kanoulas. *On the Robustness of LLM-Based Dense Retrievers: A Systematic Analysis of Generalizability and Stability*. Under review at ACM Transactions on Information Systems (**TOIS**). Submitted: 08 April 2026. [C] [Paper]
- **Yongkang Li**, Panagiotis Eustratiadis, and Evangelos Kanoulas. Spectral Tempering for Efficient Embedding Compression in Dense Passage Retrieval. In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C] [Paper]
- Arne Eichholtz, **Yongkang Li***, Jutte Vijverberg, Tobias Groot and Mohammad Aliannejadi. *Hypencoder Revisited: Reproducibility and Analysis of Non-Linear Scoring for First-Stage Retrieval.*In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C]
- **Yongkang Li**, Panagiotis Eustratiadis, Simon Lupart, and Evangelos Kanoulas. *Unsupervised Corpus Poisoning Attacks in Continuous Space for Dense Retrieval*. In the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2025**). [C] [Paper]
- **Yongkang Li**, Panagiotis Eustratiadis, and Evangelos Kanoulas. *Reproducing HotFlip for Corpus Poisoning Attacks in Dense Retrieval*. In the 47th European Conference on Information Retrieval (**ECIR'2025**). **Best Reproducibility Paper Award**. [C] [Paper]
- Dominykas Seputis, **Yongkang Li***, Karsten Langerak, and Serghei Mihailov. *Rethinking the Privacy of Text Embeddings: A Reproducibility Study of "Text Embeddings Reveal (Almost) As Much As Text"*. In the 19th ACM Conference on Recommender Systems **(RecSys'2025)**. [C] [Paper]
- **Yongkang Li**, Zipei Fan, and Xuan Song. *Heterogeneous Hyperbolic Hypergraph Neural Network for Friend Recommendation in Location-based Social Network*. ACM Transactions on Knowledge Discovery from Data (**TKDD**), 2024. [J] [Paper]
- **Yongkang Li**, Zipei Fan, Jixiao Zhang, Dengheng Shi, Tianqi Xu, Du Yin, Jinliang Deng, and Xuan Song. *Heterogeneous Hypergraph Neural Network for Friend Recommendation with Human Mobility*. In the 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**), 2022. [C] [Paper]
- **Yongkang Li**, Zipei Fan, Du Yin, Renhe Jiang, Jinliang Deng, and Xuan Song. *HMGCL: Heterogeneous Multigraph Contrastive Learning for LBSN Friend Recommendation*. World Wide Web (**WWWJ**), 2022. [J] [Paper]

Papers - Co-Author

- Kidist Amde Mekonnen, **Yongkang Li**, Yubao Tang, Simon Lupart and Maarten de Rijke. *Lost in Decoding? Reproducing and Stress-Testing the Look-Ahead Prior in Generative Retrieval*. In the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2026**). [C]
- Yuxi Lin, **Yongkang Li**, Jie Xing, and Zipei Fan. *Multifaceted Scenario-Aware Hypergraph Learning for Next POI Recommendation*. In the 40th AAAI Conference on Artificial Intelligence (**AAAI'2026**). [C] [Paper]
- Peter Atandoh, **Yongkang Li**, Weikang Guo, Jingyu Guo, Jie Zou. *Less is More: Adaptive Prompt Compression and Exemplar Selection for Efficient Few-Shot Sentiment Analysis*. Expert Systems with Applications (**ESWA**). [J] [Paper]
- Jixiao Zhang, **Yongkang Li**, Ruotong Zou, Jingyuan Zhang, Renhe Jiang, Zipei Fan, and Xuan Song. *Hyper-relational Knowledge Graph Neural Network for Next POI Recommendation*. World Wide Web (**WWWJ**), 2024. [J] [Paper]

Self-evaluation

I am intellectually curious and proactive in exploring new ideas. I work effectively under pressure, take ownership of deliverables end-to-end, and collaborate well in teams with a strong sense of responsibility.